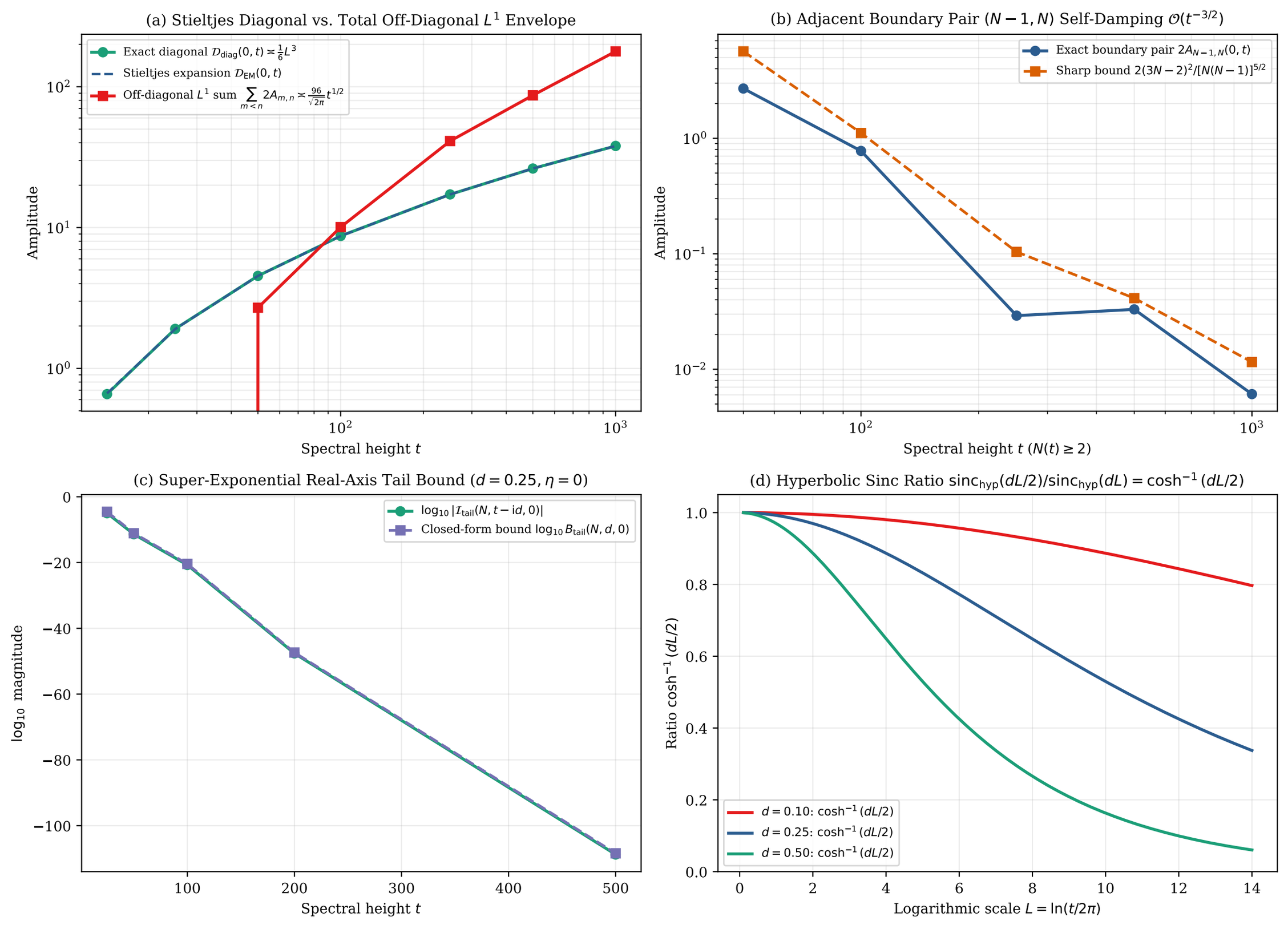
(a) Stieltjes Diagonal vs. Total Off-Diagonal L^1 Envelope
Exact diagonal $\mathcal{D}_{\mathrm{diag}}(0,t) \asymp \frac{1}{6}L^3$
Stieltjes expansion $\mathcal{D}_{\mathrm{EM}}(0,t)$
Off-diagonal L^1 sum $\sum_{m<n} 2A_{m,n} \asymp \frac{96}{\sqrt{2\pi}} t^{1/2}$
Amplitude
Spectral height t
(b) Adjacent Boundary Pair $(N-1, N)$ Self-Damping $\mathcal{O}(t^{-3/2})$
Exact boundary pair $2A_{N-1,N}(0,t)$
Sharp bound $2(3N-2)^2/[N(N-1)]^{5/2}$
Amplitude
Spectral height t $(N(t) \geq 2)$
(c) Super-Exponential Real-Axis Tail Bound $(d = 0.25, \eta = 0)$
$\log_{10} |\mathcal{I}_{\mathrm{tail}}(N, t - \mathrm{i}d, 0)|$
Closed-form bound $\log_{10} B_{\mathrm{tail}}(N, d, 0)$
$\log_{10}$ magnitude
Spectral height t
(d) Hyperbolic Sinc Ratio $\mathrm{sinc}_{\mathrm{hyp}}(dL/2)/\mathrm{sinc}_{\mathrm{hyp}}(dL) = \cosh^{-1}(dL/2)$
$d = 0.10$: $\cosh^{-1}(dL/2)$
$d = 0.25$: $\cosh^{-1}(dL/2)$
$d = 0.50$: $\cosh^{-1}(dL/2)$
Ratio $\cosh^{-1}(dL/2)$
Logarithmic scale $L = \ln(t/2\pi)$